

1 • DEFINITIONS & STRUCTURE

WILLIS' DEFINITION — LEARN IT VERBATIM "A neoplasm is an abnormal mass of tissue, the growth of which exceeds and is uncoordinated with that of the normal tissue, and persists in the same excessive manner after cessation of the stimulus which evoked the change."

Two components of every tumour:

PARENCHYMA	the neoplastic cells — they decide the	name and behaviour	+	STROMA	connective tissue + blood vessels — decides the	growth and consistency
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- **Desmoplasia / scirrhous** = abundant fibrous stroma → a **stony-hard** tumour
- **Medullary** = little stroma → soft, fleshy tumour
- **Oncology** = study of tumours · **cancer** = any malignant tumour

2 • NOMENCLATURE ★★

Tissue of origin	Benign	Malignant
Surface epithelium	Papilloma	Squamous cell carcinoma
Glandular epithelium	Adenoma	Adenocarcinoma
Fibrous tissue	Fibroma	Fibrosarcoma
Fat	Lipoma	Liposarcoma
Bone	Osteoma	Osteosarcoma
Cartilage	Chondroma	Chondrosarcoma
Blood vessel	Haemangioma	Haemangiosarcoma
Smooth muscle	Leiomyoma	Leiomyosarcoma
Skeletal muscle	Rhabdomyoma	Rhabdomyosarcoma
Melanocyte	Melanocytoma	Melanoma

THE "-OMA" EXCEPTIONS — MALIGNANT DESPITE THE NAME ★
Lymphoma · **Melanoma** · **Seminoma** · **Mesothelioma** · **Hepatoma** · **Thymoma (locally invasive)**. Also remember the **non-neoplastic "-omas"**: **haematoma**, **granuloma**, **hamartoma**.

THREE LOOK-ALIKE TERMS **Hamartoma** — a disorganised mass of tissue native to the site. **Choristoma** — normal tissue in the **WRONG place** (ectopic). **Teratoma** — from germ cells, contains more than one germ layer (hair, teeth, bone in an ovarian mass).

Rule: malignant tumour of **epithelium** = **CARCINOMA**; of **mesenchyme** = **SARCOMA**. A **carcinosarcoma** has both.

3 • BENIGN VS MALIGNANT ★★★

Feature	Benign	Malignant
Growth rate	Slow	Rapid
Mode of growth	Expansile , pushes tissue aside	Infiltrative / invasive
Capsule	Present , well demarcated	Absent , ill-defined margin
Metastasis	NEVER	Characteristic
Differentiation	Well differentiated, resembles parent	Poor — anaplasia
Mitoses	Few, normal	Many, abnormal/tripolar
Nucleus	Normal, N:C ≈ 1:4–1:6	Hyperchromatic , N:C → 1:1, big nucleoli
Pleomorphism	Absent	Marked (variable size & shape)
Necrosis & haemorrhage	Rare	Common (outgrows blood supply)
Recurrence after removal	Rare	Frequent
Effect on host	Usually local (pressure)	Cachexia , often fatal

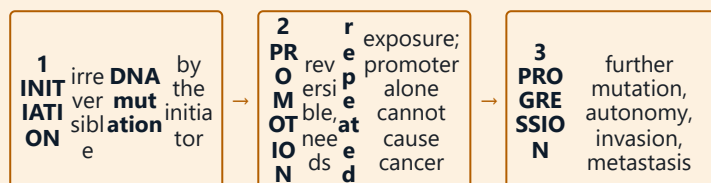
THE SINGLE DECISIVE CRITERION METASTASIS — its presence proves malignancy. But **absence does not prove benignity**, because some malignant tumours (basal cell carcinoma, most brain tumours) rarely metastasise.

4 • ANAPLASIA — THE MICROSCOPIC CHECKLIST

- **Pleomorphism** — variation in cell and nuclear size and shape
- **Hyperchromasia** — dark nuclei from ↑ DNA
- **High nucleo-cytoplasmic ratio** — up to 1:1
- **Prominent, multiple, irregular nucleoli**
- **Abnormal mitotic figures** — tripolar, quadripolar, ring; ↑ mitotic index
- **Tumour giant cells** with single huge or multiple nuclei
- **Loss of polarity** — disorderly, chaotic arrangement
- **Ischaemic necrosis** in the centre of the mass

THE PRECANCER LADDER Hyperplasia → Metaplasia → **DYSPLASIA (reversible, pre-neoplastic)** → **Carcinoma in situ (full-thickness change, basement membrane INTACT)** → **Invasive carcinoma (basement membrane BREACHED)**.

5 • AETIOLOGY & CARCINOGENESIS ★★

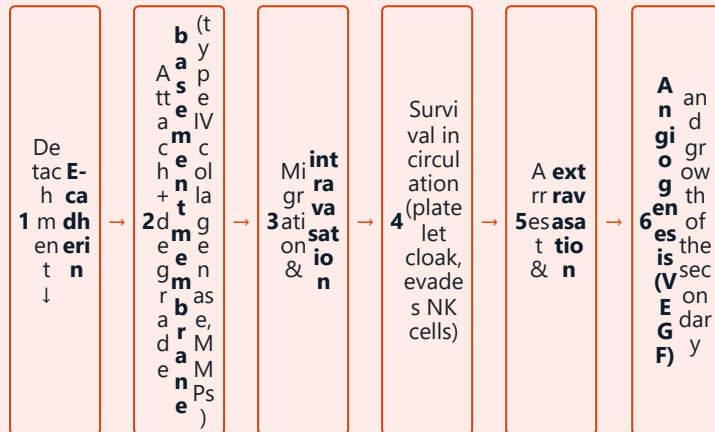


Class	Agents in veterinary practice
Chemical	Aflatoxin B₁ → hepatocellular carcinoma · bracken fern (ptaquiloside) → bladder and upper GI tumours · nitrosamines · tar, soot, benzpyrene · cycad, ragwort
Physical	Ultraviolet light → SCC of eyelid and third eyelid in Hereford cattle , ear tips of white cats · ionising radiation · chronic irritation
Viral (oncogenic) ★	Bovine papillomavirus → warts, equine sarcoid , and with bracken fern → enzootic haematuria Marek's disease herpesvirus → T-cell lymphoma of chickens Avian leukosis/sarcoma virus → B-cell lymphoid leukosis; Rous sarcoma virus — the first tumour virus (Peyton Rous) Bovine leukaemia virus → enzootic bovine leukosis (adult multicentric) FeLV → feline lymphoma & leukaemia · Shope fibroma/papilloma of rabbits
Hormonal	Oestrogen → mammary tumours of the bitch; testosterone → perianal (hepatoid) gland adenoma
Parasitic	Spirocerca lupi → oesophageal fibrosarcoma/osteosarcoma in dogs

GENES **Oncogenes** (accelerator, gain of function) — **ras, myc, erbB, src**; the normal form is a **proto-oncogene**. **Tumour suppressor genes** (brake, loss of function) — **p53 "guardian of the genome"**, **Rb**. **Anti-apoptotic** — **bcl-2**. Plus **DNA-repair genes** and re-activated **telomerase**.

6 • SPREAD OF NEOPLASMS ★★★

Route	Typical of
1 Direct / local invasion	All malignancies — the commonest
2 Lymphatic	CARCINOMAS — to the regional lymph node first
3 Haematogenous (blood)	SARCOMAS — veins are thin-walled; first filter is the lung (systemic veins) or liver (portal drainage)
4 Transcoelomic	Seeding across body cavities — ovarian and intestinal carcinoma → peritoneal carcinomatosis with ascites
5 Implantation / iatrogenic	Along a needle or surgical track
6 Transplantation ★	Canine transmissible venereal tumour (TVT) — the living tumour cell itself is transferred at coitus; the cells carry ~59 chromosomes instead of the normal canine 78 . Also devil facial tumour disease

Metastatic cascade — the order matters


TWO FACTS EXAMINERS LOVE ① **Benign tumours never metastasise.** ② A tumour cannot grow beyond about **1–2 mm** without **angiogenesis**.

7 • TUMOUR IMMUNITY

- **Tumour-specific antigen (TSA)** — found only on tumour cells; **tumour-associated antigen (TAA)** — also on some normal cells
- **Effectors:** **cytotoxic T lymphocytes (CD8, the main one)** · **NK cells** (kill MHC-I-negative cells) · activated macrophages · antibody with ADCC
- **Immunosurveillance** — the immune system continuously removes transformed cells; tumours appear when it fails, hence more tumours in **immunosuppressed** animals
- **Escape mechanisms:** loss of MHC class I · antigen shedding & **blocking antibody** · **T-reg** and TGF-β mediated suppression · tumour cells outgrow the response (sneaking through) · antigenic modulation

CLASSIC PROOF IN VETERINARY SCIENCE **TVT regresses spontaneously** in an immunocompetent dog — the standard teaching example of effective tumour immunity.

8 • EFFECTS ON THE HOST & PARANEOPLASTIC SYNDROMES

Local — pressure atrophy, obstruction (gut, airway, urethra), ulceration and haemorrhage, secondary infection, pain, pathological fracture.

Systemic — **CACHEXIA** (mediated largely by **TNF-α**, "cachectin", plus IL-1 and IL-6), anaemia, fever, DIC, thrombo-embolism.

Paraneoplastic syndrome	Tumour behind it
Hypercalcaemia of malignancy (PTH-related protein)	Lymphoma and anal sac apocrine gland adenocarcinoma of the dog
Feminisation — gynaecomastia, symmetric alopecia, pendulous prepuce, bone-marrow aplasia	SERTOLI CELL TUMOUR (hyperoestrogenism), often in a cryptorchid testis
Hypoglycaemia	Insulinoma , large hepatic tumours
Polycythaemia	Renal tumour (erythropoietin)
Hyperhistaminaemia → gastric ulcer	Mast cell tumour of the dog

9 • DIAGNOSIS • GRADING • STAGING ★★★

Diagnostic methods — clinical exam and imaging · **fine-needle aspiration cytology** (quick, cheap, no architecture) · **impression smear** · **BIOPSY + histopathology = the GOLD STANDARD** (incisional, excisional, needle-core, punch, endoscopic) · **immunohistochemistry** (cytokeratin → epithelial; vimentin → mesenchymal; CD3 → T cell; CD79a/CD20 → B cell) · electron microscopy · flow cytometry.

	GRADING	STAGING
Based on	MICROSCOPIC appearance	CLINICAL / anatomical EXTENT
Criteria	Degree of differentiation , mitotic index , necrosis, pleomorphism, invasion	T = size/extent of primary · N = regional lymph nodes · M = distant Metastasis
Scale	Grade I well differentiated → III poorly differentiated/anaplastic	Stage 0–IV
Value	Tells how the tumour behaves	Better predictor of survival ; guides therapy

REMEMBER BY THE WORDS Grading = **G**lass slide (microscope). Staging = **S**pread (how far it has gone).

10 • EPITHELIAL TUMOURS

Tumour	Key point
Papilloma (warts)	Papillomavirus , species-specific; cauliflower-like, fibrovascular core with hyperkeratotic epithelium; usually regresses spontaneously ; teat and penile warts in cattle
Squamous cell carcinoma ★	Sunlight + unpigmented skin. Cattle — eye/third eyelid (" cancer eye " of Hereford); cat — ear tip and nose of white cats; horse — penis, vulva, eye; dog — nail bed. Micro: invading nests with KERATIN (epithelial) PEARLS and intercellular bridges
Basal cell tumour	Cat and dog, head and neck; locally invasive, rarely metastasises
Adenoma / adenocarcinoma	Perianal (hepatoid) gland adenoma — entire male dog, testosterone-dependent ; thyroid, hepatic, renal, intestinal
Mammary tumours	Commonest tumour of the entire bitch ; mixed mammary tumour (epithelial + cartilage/bone) is characteristic of the dog; roughly 50% are malignant in the bitch, but ~85–90% in the queen . Early spaying is strongly protective
Transitional cell carcinoma	Urinary bladder; trigone; associated with bracken fern in cattle

11 • MESENCHYMAL & ROUND CELL TUMOURS ★★

Tumour	Key point
Fibroma / fibrosarcoma	Fibrosarcoma at vaccination sites in cats ; interlacing "herring-bone" bundles
Lipoma	Soft, well-encapsulated, yellow, greasy; floats in formalin
OSTEOSARCOMA ★	Commonest primary bone tumour ; large-breed dogs, metaphysis of long bones — " away from the elbow, toward the knee " (distal radius, proximal humerus, distal femur, proximal tibia). Early pulmonary metastasis ; produces tumour osteoid
Haemangiosarcoma	SPLEEN and right atrium of the dog (German Shepherd); ruptures → haemoabdomen and sudden collapse
Equine sarcoid	Commonest skin tumour of the horse ; bovine papillomavirus 1 & 2 ; locally aggressive, recurs, does not metastasise
Melanoma	Grey horses — perineum, tail base, parotid region; usually slow but may become malignant. In dogs, oral melanoma is highly malignant ; skin melanoma often benign

THE FIVE ROUND CELL TUMOURS — A FAVOURITE QUESTION 1

Lymphoma · **2 Mast cell tumour** (metachromatic granules with **toluidine blue/Giemsa**; ↑ eosinophils) · **3 Histiocytoma** (young dog, "button tumour", **regresses on its own**) · **4 Plasmacytoma** · **5 Transmissible venereal tumour (TVT)** (external genitalia, **vacuolated cytoplasm**, transplanted cells).

Lymphosarcoma in cattle — **4 forms**: **adult/enzootic (BLV)** multicentric · **juvenile (calf)** · **thymic** (young adult) · **cutaneous**. The first is virus-associated; the other three are **sporadic** and not BLV-related.

Gonadal tumours: **Sertoli cell tumour** → feminisation; **seminoma** → from germ cells; **interstitial (Leydig) cell tumour** → usually incidental; **granulosa cell tumour** of the ovary → persistent oestrus or anoestrus, may make inhibin.

12 • ANTICOAGULANTS ★★★ (GUARANTEED QUESTION)

Anticoagulant	Mechanism	Use — and the trap
EDTA (di-potassium salt)	Chelates calcium	THE anticoagulant for HAEMATOLOGY — best preservation of cell morphology. ~ 1–2 mg per ml of blood . Excess EDTA shrinks cells → falsely low PCV . Never use for calcium estimation
Heparin (lithium/sodium)	Activates antithrombin III , inhibits thrombin	Best for plasma biochemistry, blood gas and osmotic fragility . ~ 10–20 IU per ml . Poor for smears — gives a blue background with Romanowsky stains and clumps leucocytes
Trisodium citrate	Chelates calcium, reversible	1 part 3.2% citrate : 9 parts blood for coagulation tests (PT, APTT) and for blood transfusion ; 1 : 4 for ESR (Westergren)
Double oxalate (ammonium + potassium, Wintrobe's)	Precipitates calcium	PCV and ESR only; crenates cells, unsuitable for smears
Sodium fluoride	Inhibits glycolysis (enolase)	GLUCOSE estimation — the only correct choice; usually combined with potassium oxalate

GOLDEN RULE The **blood smear must be made within minutes, ideally from FRESH blood without any anticoagulant**. If that is not possible, use EDTA and make the smear within **1 hour**.

13 • HB, PCV, TEC AND THE THREE INDICES ★★★

- Haemoglobin** — **cyanmethaemoglobin (HiCN) method with Drabkin's solution is the international REFERENCE method**, read at **540 nm**. **Sahli's acid-haematin** is the visual, less accurate field method. Oxyhaemoglobin method is also used.
- PCV / haematocrit** — **microhaematocrit** capillary spun at high speed for ~5 min (the routine method), or the **Wintrobe** tube. The **buffy coat** between plasma and red cells holds the leucocytes and platelets.
- TEC and TLC** — **Neubauer counting chamber**. **RBC diluting fluid = Hayem's**; **WBC diluting fluid = Turk's** (glacial acetic acid **lyses the red cells**, gentian violet stains the leucocyte nuclei).

Index	Formula	Unit	Tells you
MCV	$\text{PCV (\%)} \times 10 \div \text{RBC (millions/\mu l)}$	fl	Cell SIZE — macro / normo / microcytic
MCH	$\text{Hb (g/dl)} \times 10 \div \text{RBC (millions/\mu l)}$	pg	Hb weight per cell
MCHC	$\text{Hb (g/dl)} \times 100 \div \text{PCV (\%)}$	g/dl (%)	Hb CONCENTRATION — normo / hypochromic

WHICH INDEX NEEDS WHICH VALUE ★ **MCV needs PCV and RBC**. **MCH needs Hb and RBC**. **MCHC needs Hb and PCV** — it is the **only index that does not use the red cell count**, so it is the most reliable when the counter is unreliable. Hypochromia is a **low MCHC**.

ESR Westergren (200 mm tube, citrate 1:4) and **Wintrobe** (100 mm tube) methods. ESR is **fastest in the HORSE** (marked rouleaux) and **very slow in cattle, sheep and goats**. It rises with ↑ fibrinogen and globulin (inflammation, pregnancy) and falls with polycythaemia and dehydration.

14 • TLC, DLC AND THE LEUCOGRAM

ABSOLUTE COUNT Absolute count = (percentage of that cell × TLC) ÷ 100. Always interpret **absolute**, never percentage — a "normal 30% lymphocytes" can still be lymphopenia if the TLC is low.

Species group Predominant leucocyte

Dog, cat, horse **NEUTROPHIL**

Cattle, sheep, goat **LYMPHOCYTE**

Pig Roughly equal / lymphocyte

Birds, reptiles, rabbits **HETEROPHIL** replaces the neutrophil (rod-shaped red granules); lymphocyte usually predominates in birds

Stains — the **Romanowsky** group: **Leishman, Giemsa, Wright, May-Grünwald** (methylene blue + eosin in methanol). Counting method: **battlement / cross-sectional, 100 cells** in the body-and-tail of the smear.

Change Meaning

Left shift, regenerative Immature (band) cells present but **mature > immature** — a good response

Left shift, DEGENERATIVE **Immature > mature**, usually with leucopenia — **overwhelming infection, poor prognosis**

Right shift **Hypersegmented** neutrophils — old cells, corticosteroid or chronic disease

Toxic change **Döhle bodies**, cytoplasmic basophilia, vacuolation, toxic granulation — severe bacterial toxemia

Stress leucogram **Neutrophilia + lymphopenia + eosinopenia + monocytosis**

Eosinophilia **Parasites and allergy**

Leukaemoid reaction Very high TLC that mimics leukaemia but is **reactive** — e.g. pyometra, abscess

15 • BLOOD SMEAR — WHAT YOU MUST RECOGNISE ★★**Finding Interpretation**

Polychromasia, anisocytosis, reticulocytes, nucleated RBC **REGENERATIVE** response — active bone marrow

Howell-Jolly body Small round **nuclear remnant**; regeneration, or after splenectomy

HEINZ BODY ★ **Denatured haemoglobin** at the cell margin; **oxidative injury** — onion/*Brassica*, kale, phenothiazine, **copper toxicity**, paracetamol in cats. Stain with **new methylene blue**; almost invisible on Giemsa

Basophilic stippling **LEAD POISONING** (with nucleated RBCs and no anaemia) — also normal regeneration in ruminants

Rouleaux Stacked coins — **normal in the horse**, and in hyperglobulinaemia

Schistocytes Fragmented cells — **DIC**, haemangiosarcoma

Spherocytes Small, dark, no central pallor — **immune-mediated haemolysis** (best seen in the dog)

Blood parasite Appearance

Babesia bigemina **Large PEAR-SHAPED** pairs at an acute angle inside the RBC

Theileria **Koch's blue bodies (schizonts) in lymphocytes** of the node; small rod/comma piroplasms in RBCs

Anaplasma marginale Small dense dot at the **MARGIN** of the RBC, no cytoplasm

Trypanosoma evansi **Extracellular** motile flagellate **between** the cells (surra)

16 • URINALYSIS ★★

PHYSICAL — volume, colour, transparency, odour, **specific gravity** (refractometer).

THE ONE URINE NUMBER TO MEMORISE ★ ISOSTHENURIA = USG 1.008–1.012, the same as plasma ultrafiltrate. Persistent isosthenuria in an azotaemic animal means the kidney **has lost its concentrating ability** — chronic renal failure. Carnivores normally concentrate well above this (dog often > 1.025, cat higher still).

CHEMICAL**Test Meaning**

pH **Herbivores ALKALINE** (≈7.5–8.5) · **carnivores ACIDIC** (≈5.5–7.0). Milk-fed calves are acidic. Alkaline urine favours **struvite** stones

Protein Glomerular damage, inflammation. Interpret with the **urine protein : creatinine ratio**; dipstick gives **false positives in alkaline urine**

Glucose Appears only above the **renal threshold** — diabetes mellitus, stress hyperglycaemia (especially **cats**), or tubular damage

Ketones **Bovine ketosis, ovine pregnancy toxemia**, diabetic ketoacidosis, starvation. Rothera's test

Bilirubin Only the **conjugated (direct)** form appears; small amounts are **normal in concentrated dog urine**, never normal in the cat

Blood / Hb / Mb **Haematuria** — intact RBCs, sediment shows red cells, spin down and the supernatant clears. **Haemoglobinuria** — supernatant stays red. **Myoglobinuria** — plasma stays clear

MICROSCOPIC (sediment)**Structure Meaning**

Casts ★ Formed in the **distal tubule and collecting duct** from **Tamm-Horsfall protein**; they prove the lesion is **renal**. **Hyaline** = protein only, mild · **granular** = degenerating cells, tubular injury · **cellular** (RBC / WBC / epithelial) · **waxy** = chronic, long-standing · **fatty** = diabetes, cat "Coffin lid" prism — alkaline urine

Struvite crystal

Calcium oxalate Dihydrate = "envelope"; monohydrate = "dumb-bell/picket fence" — **ethylene glycol poisoning**

Ammonium biurate "Thorn apple" — liver failure, portosystemic shunt

Cystine Flat **hexagonal** plates — always abnormal

17 • BIOPSY & CYTOLOGY

- **Biopsy types:** **excisional** (whole lesion out — diagnosis + treatment) • **incisional/wedge** • **needle-core (Tru-cut)** • **punch** • endoscopic • bone-marrow aspirate
- **Cytology techniques:** **fine-needle aspiration** • impression (touch) smear • scraping • swab • **EXFOLIATIVE cytology** of fluids — pleural, peritoneal, CSF, tracheal wash, vaginal, urine
- **Fixation:** **air-dry** for Romanowsky stains; **wet-fix immediately in 95% ethanol** for Papanicolaou and H&E
- **Cytological criteria of malignancy** — **anisocytosis** and **anisokaryosis**, high N:C ratio, multiple prominent angular nucleoli, coarse chromatin, abnormal mitoses, multinucleation, cannibalism

CYTOLOGY VS HISTOPATHOLOGY Cytology is **rapid, cheap, minimally invasive**, but it **cannot show tissue ARCHITECTURE** and therefore cannot prove **INVASION**. Histopathology remains the **gold standard** for the final diagnosis.

18 • NECROPSY — OBJECTIVES & PROCEDURE ★★

Objectives — establish the **cause of death** • confirm or correct the clinical diagnosis • detect disease in the rest of the herd (surveillance) • **insurance and legal** requirements • check response to treatment • teaching and research.

Pre-necropsy rules — obtain full **history** and **written owner consent** (in a veterolegal case, a police or magistrate **requisition** replaces it) • perform **as soon as possible after death**, in **daylight** • wear protective clothing, remember **zoonoses** (anthrax, rabies, brucellosis, TB, glanders, leptospirosis) • arrange disinfectant and safe carcass disposal.

THE ANTHRAX RULE ★★★ Never open a ruminant found suddenly dead with unclotted tarry blood at the orifices. Take a **peripheral blood smear** first and stain with **polychrome methylene blue (McFadyean's reaction)**. Only if anthrax is excluded may the carcass be opened. Dispose by **deep burial with quicklime** or **incineration**.

Item	Standard practice
Position	Ruminants and horses — LEFT lateral recumbency (left side down) so the rumen lies down and the right-side viscera are exposed. Dog, cat, pig, poultry — DORSAL recumbency
Techniques	Virchow — organs removed one by one • Rokitansky — dissected in situ • Ghon — removed en bloc by systems • Letulle — all viscera removed en masse
Wild animals	Chemical immobilisation records, photograph everything, check for poaching wounds, collect samples for toxicology, follow forest-department protocol

19 • POST-MORTEM CHANGES ★★★

Change	What it is
Algor mortis	Cooling of the body — slower in a fat animal and in Indian summer heat
RIGOR MORTIS	ATP depletion locks actin to myosin. Begins in ~ 1–6 h , complete by about 12 h , passes off in 24–48 h as autolysis proceeds; faster onset and faster passing in hot weather . Begins in the heart and small muscles of the head . ABSENT or very poor after violent exertion, in anthrax and in septicaemia
Livor mortis / hypostatic congestion	Blood settles in the down side ; uniform, gravity-dependent
Post-mortem clotting	"Chicken fat" and "currant jelly" clots — shiny, rubbery, unattached, a cast of the vessel
Haemoglobin imbibition	Red staining of endocardium and intima
Bile imbibition	Green-yellow staining of tissue around the gall bladder
Pseudomelanosis	Blue-green to black discolouration where gut bacteria make H₂S which combines with iron — abdominal wall and intestines
Post-mortem emphysema & bloating	Gas from putrefactive bacteria; may rupture the stomach or diaphragm
Autolysis	Self-digestion — earliest in gut mucosa, pancreas, adrenal medulla, gall bladder, brain
Corneal clouding, lens opacity	Useful rough guide to the time since death

20 • SPECIMEN COLLECTION & DISPATCH ★★★

Purpose	Preservative & rule
Histopathology	10% NEUTRAL BUFFERED FORMALIN , fixative : tissue volume 10 : 1 , tissue slice not thicker than about 0.5–1 cm , include the junction of normal and abnormal tissue
Brain (rabies)	50% buffered glycerol saline for transport, or chilled unpreserved for FAT (the standard test) and mouse inoculation; take hippocampus (Ammon's horn), cerebellum and brain stem . Half the brain in formalin for histology
Eye	Bouin's or Davidson's fluid — plain formalin causes retinal detachment artefact
Bacteriology	Fresh, sterile, NO preservative , chilled on ice, sent within 24 h; sear the surface before sampling
Virology	Fresh chilled or 50% glycerol saline , transported on ice / dry ice
TOXICOLOGY ★	NO chemical preservative — send fresh or frozen ; if unavoidable, use saturated salt solution and send a control sample of the salt. Send liver, kidney, stomach content, ingesta, suspect feed/water , a generous quantity (about 100–500 g)
Parasitology	Helminths in 70% alcohol or 10% formalin; faeces fresh and chilled
Blood parasites	Thin peripheral blood smear , air-dried, from the ear-tip capillary
Packing: leak-proof primary container → absorbent material → sturdy secondary container → outer box with ice packs; label each container and enclose a forwarding letter with full history, treatment given, and the tests requested .	

21 • PM REPORT • VETEROLEGAL NECROPSY & WOUNDS ★★**The post-mortem report — three parts**

- 1. Preliminary** — identification of the animal (species, breed, age, sex, marks, tag), owner, history, date, time and place of death and of the necropsy, condition of the carcass
- 2. Gross findings** — organ by organ, **DESCRIBE only**; state size, shape, colour, consistency, distribution. **Do not write a diagnosis in this section**
- 3. Conclusion** — **morphological diagnosis, aetiological diagnosis, cause of death**, samples sent, remarks, signature with registration number

VETEROLEGAL NECROPSY — THE RULES Carried out only on a **written requisition** from the police or a magistrate · establish the **identity** of the animal beyond doubt · **photograph** everything · perform in the presence of the investigating officer · samples are **sealed with the officer's seal** and the **chain of custody** is documented · write **in ink, no overwriting or erasure**, correct by scoring through and initialling · the report goes to the **requisitioning authority, not the owner** · retain a copy.

Wound	Distinguishing feature
Incised	Sharp edge; LENGTH greater than DEPTH ; clean margins, bleeds freely
Stab / punctured	DEPTH greater than LENGTH ; small external opening
Lacerated	Blunt force; irregular torn ragged margins , tissue bridges, hair driven in
Contused (bruise)	Blunt force with skin intact ; colour changes with age of the bruise
Abrasion	Superficial, loss of epidermis only
Firearm	Entry — small, inverted, with a collar of abrasion and possibly scorching/tattooing. Exit — larger, everted, no scorching

ANTEMORTEM VS POSTMORTEM WOUND ★★★ An **antemortem** wound shows a **VITAL REACTION** — free arterial **haemorrhage with clotting**, infiltration of blood into the surrounding tissue, **gaping and retraction of edges, inflammation**, and later signs of **healing**. A **postmortem** wound shows none of these — the edges do not gape, blood merely oozes and does not clot into the tissue.

22 • BLANK-FILL DRILL — UNIT 3

- The two components of a tumour are **parenchyma** and **stroma**
- Abundant fibrous stroma makes a tumour **scirrhous** (desmoplastic)
- Malignant tumour of epithelium is a **carcinoma**; of mesenchyme a **sarcoma**
- Lymphoma, melanoma, seminoma and mesothelioma are **malignant** despite the "-oma"
- A teratoma contains more than one **germ layer**
- The single criterion that proves malignancy is **metastasis**
- Carcinomas spread mainly by **lymphatics**; sarcomas by **blood**
- TVT spreads by **transplantation** of living tumour cells and has about **59** chromosomes
- Loss of **E-cadherin** allows tumour cells to detach
- A tumour cannot exceed **1–2 mm** without **angiogenesis**
- p53 is called the **guardian of the genome**

- Aflatoxin B₁ causes **hepatocellular carcinoma**
- Marek's disease virus causes a **T-cell lymphoma** of chickens
- The first tumour virus discovered was the **Rous sarcoma virus**
- Cancer eye of Hereford cattle is a **squamous cell carcinoma** caused by **ultraviolet light**
- Keratin (epithelial) pearls are seen in **squamous cell carcinoma**
- Osteosarcoma occurs **away from the elbow and toward the knee**
- Splenic haemangiosarcoma causes sudden **haemoabdomen**
- Equine sarcoid is caused by **bovine papillomavirus** and does **not** metastasise
- The five round cell tumours are lymphoma, **mast cell tumour**, histiocytoma, plasmacytoma and **TVT**
- Sertoli cell tumour causes **feminisation** and bone-marrow **aplasia**
- Hypercalcaemia of malignancy in dogs is classic of **lymphoma** and **anal sac adenocarcinoma**
- Grading is **microscopic**; staging uses the **TNM** system
- EDTA works by **chelating calcium** and is the anticoagulant of choice for **haematology**
- Heparin acts through **antithrombin III** and is best for **plasma biochemistry**
- Citrate is used at **1:9** for coagulation tests and **1:4** for ESR
- Sodium fluoride is used for **glucose** estimation because it blocks **glycolysis**
- The reference method for haemoglobin is **cyanmethaemoglobin**, read at **540** nm
- WBC diluting fluid is **Turk's**; RBC diluting fluid is **Hayem's**
- MCHC = $\text{Hb} \times 100 \div \text{PCV}$ — the only index that does not need the **red cell count**
- ESR is fastest in the **horse** and slowest in **cattle**
- Cattle and sheep normally have a predominance of **lymphocytes**
- A degenerative left shift means **immature exceed mature** neutrophils and carries a **poor** prognosis
- Heinz bodies indicate **oxidative** injury and are stained by **new methylene blue**
- Basophilic stippling with nucleated RBCs suggests **lead** poisoning
- Anaplasma appears at the **margin** of the red cell; Trypanosoma lies **outside** the cells
- Koch's blue bodies are the schizonts of **Theileria** in **lymphocytes**
- Isosthenuria is a urine specific gravity of **1.008–1.012**
- Herbivore urine is normally **alkaline**; carnivore urine **acidic**
- Urinary casts form in the **distal tubule and collecting duct** from **Tamm-Horsfall** protein
- Coffin-lid crystals are **struvite**; envelope crystals are **calcium oxalate**
- Ruminants are placed in **left** lateral recumbency for necropsy
- Rigor mortis is due to depletion of **ATP** and is absent in **anthrax**
- Pseudomelanosis is caused by **hydrogen sulphide** combining with iron
- Toxicology samples must be sent with **no chemical preservative**
- An antemortem wound is proved by the presence of a **vital reaction**
- In an incised wound the **length** exceeds the depth; in a stab wound the **depth** exceeds the length